Predicting Turkish Airlines Stock Prices using N-BEATS, TFT, and Informer Models in the Amid High Turkish Inflation

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Abstract. This study assesses the effectiveness of advanced deep learning-based forecasting models, namely Informer, N-BEATS, and Temporal Fusion Transformer (TFT), in forecasting the adjusted closing price of Turkish Airlines (THYAO) shares over the period 2021-2025, a period marked by prolonged inflation and macroeconomic fluctuations in Turkey. The research is based on 1,167 daily data points obtained from Yahoo Finance, where historical price differences are used to identify temporal patterns in stock price behavior. Model performance is evaluated using a rolling cross-validation method, providing a realistic and robust analysis in a turbulent and unstable market environment. Forecast accuracy is assessed by applying RMSE, MAE, and MAPE metrics. The results of the study show that the univariate Informer model has the highest accuracy and prediction stability, outperforming N-BEATS and TFT. Although N-BEATS successfully captures non-linear price dynamics well, this model shows a slight delay when sudden trend changes occur. On the other hand, the multivariate TFT model showed a higher error rate, indicating that the addition of external variables such as Open, High, Low, and Volume does not always improve forecasting results and can add noise. These findings are further supported by post-hoc ablation analysis, which reveals that some external variables have a limited impact or even reduce prediction accuracy. Overall, this study emphasizes the superiority of transformer-based univariate models for predicting stock prices in the context of highly volatile and inflation-prone emerging markets, providing important insights for investors and policymakers.

Keywords: Time Series Forecasting · Stock Price Prediction · High Inflation Economy · Informer · N-BEATS · Temporal Fusion Transformer · Rolling Cross Validation · Airline Industry · Turkish Stock Market

1 Introduction

A stock is a basic unit of investor ownership in a company in the sense of being the owner of the company [1]. The price of stocks on the stock exchange certainly fluctuates from time to time, which is certainly an attraction for investors to understand the development of stock prices on the stock exchange [2]. In theory, changes in stock prices are influenced by several macroeconomic variables, two of which are inflation and the BI Rate [2]. If the value of a country's currency falls, it indicates that other currencies have risen and strengthened, which indicates that the economy is not performing well [3].

Turkey is one of the countries that has been in the spotlight in recent years due to its inflation and highly unstable macroeconomic dynamics. Turkey even ranks fifth as the country with the highest inflation rate in the world, at 33.29% in September 2025 [4].

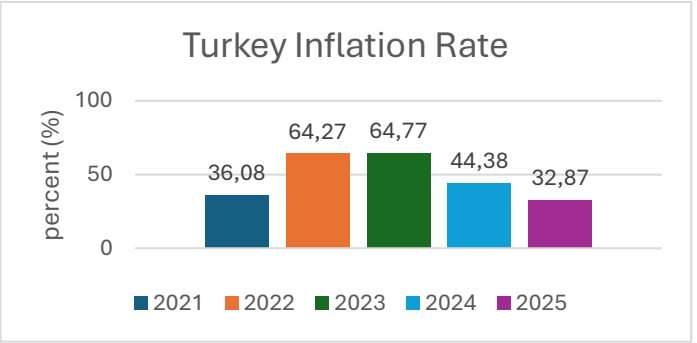


Fig. 1. *Turkey Inflation Rate (CPI)*
Source : Inflation Tool.com, 2025 [5]

Based on Figure 1, it can be seen that entering 2022, inflation rose dramatically to 64.27%, driven by the weakening of the lira, increases in global energy prices, and the government's continued low interest rate policy. These conditions had a direct impact on the financial markets, especially sectors that are sensitive to changes in exchange rates and global commodity prices. One of the most affected sectors is aviation. Turkish Airlines (THYAO) is the national airline and one of the largest and best in the world, ranking sixth globally [6]. THYAO has a cost structure that depends on exchange rates, fuel prices, and international funding, making its share price highly sensitive to macroeconomic turmoil. Thus, THYAO's share price not only reflects the company's internal performance but also serves as an indicator of Turkey's overall economic condition.

Several studies on Turkish Airlines stock price prediction have been conducted using machine learning and deep learning approaches, particularly using sequential neural network-based models such as LSTM, RNN, and GRU. The first study by Gür et al. (2024) [7] used daily THYAO data from 2010 to 2023 and compared three models, namely SVM, XGBoost, and LSTM. The results showed that LSTM was the most accurate model because it was able to capture non-linear patterns and long-term correlations in time series data.

The second study by Sezgin et al. (2025) [8] predicted the daily closing prices of THYAO and PGSUS using RNN, LSTM, and GRU optimized with Bayesian Optimization. The results show that the GRU model provides the best performance with a MAPE value of 3.05% for THYAO, which is categorized as highly accurate forecasting. This indicates that the GRU-based RNN model consistently excels at learning sequential stock price patterns.

The third study by Kalyoncu et al. (2020) [9] used only one prediction model, namely LSTM, because it is considered an advanced version of RNN that is specifically designed to handle sequential data with long-term dependency patterns. The results of the study show that LSTM is capable of providing good accuracy, in line with the findings of previous studies that place LSTM as one of the most effective methods for stock price forecasting.

Based on information obtained from Medium.com (2023) [10], LSTM learns long-term patterns but is slow to train, has high computational costs, and is prone to overfitting. RNN models are capable of capturing short-term temporal relationships and sequential data, but are prone to vanishing gradients [11]. GRU models are capable of capturing temporal dependencies and sequential data, but are limited in their ability to handle long-term dependencies and are computationally expensive [12]. Furthermore, the SVM model is resistant to overfitting in high-dimensional spaces, but has high algorithmic complexity and

is memory-intensive [13]. Finally, the XGBoost model is highly flexible and has high accuracy, but is very complex and its algorithm can still overfit [14].

From previous research, there are several research gaps that need to be addressed. Most studies still use models that focus on recurrent architectures such as LSTM, GRU, and RNN, which are less efficient at handling large data sets and less capable of capturing long-term dependencies optimally. In addition, there has been no research using Transformer-based models such as Temporal Fusion Transformer (TFT) or Informer, even though these models have shown advantages in long-horizon forecasting and can analyze complex time series patterns. Previous studies have also not explored the N-BEATS model, a very powerful pure deep learning architecture for univariate forecasting that often outperforms LSTM. Based on these gaps, this study applies three modern models, namely Temporal Fusion Transformer (TFT), Informer, and N-BEATS. These three models attempt to overcome the limitations of previous approaches that only relied on sequential architectures and did not utilize the power of transformer-based forecasting approaches.

In addition, the two models used in this study have a strong development foundation. The Temporal Fusion Transformer (TFT) was developed by Bryan Lim and Sercan Ö. Arik from Google Cloud AI in 2021 [15]. On the other hand, Informer was introduced by Haoyiet Sun and his team from Tsinghua University and Huawei Noah's Ark Lab in 2021 [16]. This additional information confirms that both models are major scientific contributions made specifically for long-sequence time-series forecasting.

In line with these issues, this study poses three main questions: (1) how effective are lag variables in reflecting past information to predict THYAO stock prices; (2) which model provides the best accuracy between N-BEATS, TFT, and Informer; and (3) how well these models are able to forecast the most recent period amid Turkey's highly volatile macroeconomic conditions. To answer these questions, this study uses time series data with the response variable being the Adjusted Close value of THYAO shares for the 2021-2025 period, while the predictor variable is formed through lag transformation to capture temporal dependence. This approach is expected to produce more accurate predictions while providing empirical contributions related to the most suitable prediction model for the highly dynamic Turkish market.

2 Dataset & Methodology

2.1 Data Source

The data set used in this study was obtained from Yahoo Finance through the historical price data page for Turkish Airlines (ticker: THYAO.IS) public shares, which are listed on the Istanbul Stock Exchange (BIST). This data set provides information on historical daily trading, including Open, High, Low, Close, Adjusted Close, and Volume. This data covers thousands of trading days, making it ideal for time series analysis and stock price prediction modeling.

This data set is automatically obtained by Yahoo Finance from official stock exchange sources and third-party financial data providers. Before publication, Yahoo Finance applies an internal cleaning, validation, and consolidation process to ensure that the information is ready for publication via its website and API endpoints. Historical data files can be downloaded directly in CSV format from the official Yahoo Finance data portal.

2.2 Research Variables

This study uses a dataset of daily stock prices for Turkish Airlines (THYAO.IS) obtained from Yahoo Finance [17]. This dataset consists of 1167 observations, which are daily trading data from January 4, 2021, to September 4, 2025. Each observation contains variables that reflect stock price and trading volume activity. These variables are used as a basis for building a price prediction model through time series analysis and machine learning methods.

Table 1. Research Variables in the THYAO Stock Price Dataset

No	Variable	Type	Explanation
1.	Date	Time	Stock trading date. Each row represents one day of trading.
2.	Open	Numeric	Opening price of the stock at the start of the trading day.
3.	High	Numeric	Highest price reached by the stock on that day.
4.	Low	Numeric	Lowest price reached by the stock on that day.
5.	Close	Numeric	Official closing price of the stock at the end of the trading day.
6.	Adj Close	Numeric	Closing price adjusted for corporate events such as dividends or stock splits.
7.	Volume	Numeric	Total number of shares traded on that day.

2.3 Algorithm

N-BEATS (Neural Basis Expansion Analysis for Interpretable Time Series Forecasting). N-BEATS was introduced by Oreshkin et al. (2020) [18] as a deep learning architecture based entirely on fully-connected neural networks designed specifically for time series forecasting without requiring specific statistical assumptions. This architecture consists of stacked blocks, where each block has a backcast and forecast mechanism. Backcast serves to reconstruct the historical part of the input, while forecast plays a role in generating predictions for the future [18]. In this case, mathematically, each block k converts the input x_k into backcast b_k and forecast y_k through a non-linear function:

$$b_k = f_{\theta_k}^{(b)}(x_k), y_k = f_{\theta_k}^{(f)}(x_k) \tag{1}$$

The backcast is then used to update the next block's input through a residual mechanism:

$$x_{k+1} = x_k - b_k \tag{2}$$

Meanwhile, the final model prediction is obtained by summing up all forecasts from each block:

$$\hat{y} = \sum_{k=1}^{K} y_k \tag{3}$$

The existence of a residual mechanism allows the model to learn time series patterns in a gradual and structured manner. In addition, to improve understanding, N-BEATS uses basis expansion through two types of bases, namely polynomial-shaped trend bases and Fourier function-based seasonality bases. In the trend block, predictions are formulated as:

$$y_k = T\alpha_k \tag{4}$$

Where T is the polynomial basis matrix and α_k is the coefficient studied by the model. Meanwhile, seasonal blocks utilize the sine and cosine basis:

$$y_k = S\beta_k \tag{5}$$

With this design, N-BEATS can produce trend and seasonal decompositions that are essentially similar to classical time series models, while retaining the flexibility of deep learning. Parameter optimization is performed using the Adam algorithm, while the model's loss function is based on historical reconstruction and prediction accuracy:

$$\mathcal{L} = \| x - \sum b_k \|^2 + \| \hat{y} - y_{\text{true}} \|^2 \tag{6}$$

In this study, N-BEATS was applied to predict the price movement of Turkish Airlines (THYAO) shares using lag features that describe autorefresive correlations over time. The block components, network depth, and training parameters were adjusted according to the characteristics of stock price volatility.

Informer. Informer is considered a supervised learning model built on the attention mechanism and consists of two main parts, namely the encoder and decoder, which function within the framework of the attention mechanism. The main difference in Informer lies in the application of sparse self-attention (probSparse attention), which is designed to select only the attention elements that contribute most to the sequence representation, thereby significantly reducing computational complexity while maintaining the quality of long-term representation [16]. Mathematically, the probSparse attention mechanism assumes that only a small portion of the attention scores are significant to the final representation. If $Q \in \mathbb{R}^{L_q \times d}$ and $K \in \mathbb{R}^{L_k \times d}$ are the query and key matrices, then traditional attention is calculated as:

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^\top}{\sqrt{d}})V \tag{7}$$

However, in probSparse attention, attention is only given to the subset that has the largest dot product value, based on probabilistic selection. This can maintain the performance of temporal representation [16]. In addition, Informer applies distilling operations and stacked encoder layers to gradually shorten the sequence length while increasing the robustness of the model in capturing complex temporal patterns. On the decoder side, the model utilizes self-attention and cross-attention on the encoder output with a causal masking approach, and applies zero-padding to the initial target so that multi-horizon predictions can be made stably and efficiently. By combining these mechanisms,

Informer is able to perform forecasting over long horizons with better speed and efficiency than conventional Transformers and LSTM/GRU-based RNN models. In this study, Informer was used to model THY stock price movement patterns using the last 64 historical steps as the input window and generating predictions along the 2025 test data horizon [16]. This model was implemented using the NeuralForecast library and optimized using the Adam algorithm, while the prediction results were evaluated using RMSE and MAPE to compare the performance of the TFT and N-BEATS models in achieving stock price forecasting accuracy.

Temporal Fusion Transformer (TFT). TFT is applied in this study as a deep learning approach specifically designed to forecast multi-horizon time series while maintaining a high level of interpretability. TFT was first introduced by (Lim et al., 2021) [19], through a publication in the International Journal of Forecasting. This model combines the strengths of Recurrent Neural Networks (RNN), multi-head self-attention, as well as gating and variable selection mechanisms, enabling it to capture short-term and long-term dynamics with high accuracy and efficiency.

The TFT architecture combines Gated Residual Networks (GRN) to control the flow of information within the network, variable selection networks to select the most relevant features from the input, static covariate encoders that take fixed features into account, LSTM encoder-decoders to model temporal dependencies, and multi-head attention layers that serve to highlight the time steps that have the most influence on the prediction. One of the key components of TFT is GRN (Gated Residual Network), which can be mathematically formulated as follows:

$$\mathrm{GRN}(x) = \mathrm{LayerNorm}(\mathrm{Gating}(x) \cdot \mathrm{ELU}(W_1 x + b_1)W_2 + b_2 + x) \tag{8}$$

Where ELU refers to the Exponential Linear Unit activation function, W_1 and W_2 are parameter weights, b_1 and b_2 are biases, $\mathrm{Gating}(x)$ is a sigmoid gate that controls how much information is passed on, and Layer Norm is layer normalization to improve stability during training. In this study, TFT is used to predict Turkish Airlines (THYAO) stock prices by utilizing lag features, time-varying known inputs, and static covariates if available [19]. The model is evaluated using the RMSE and MAPE metrics and compared with the performance of N-BEATS and Informer.

2.4 Methods

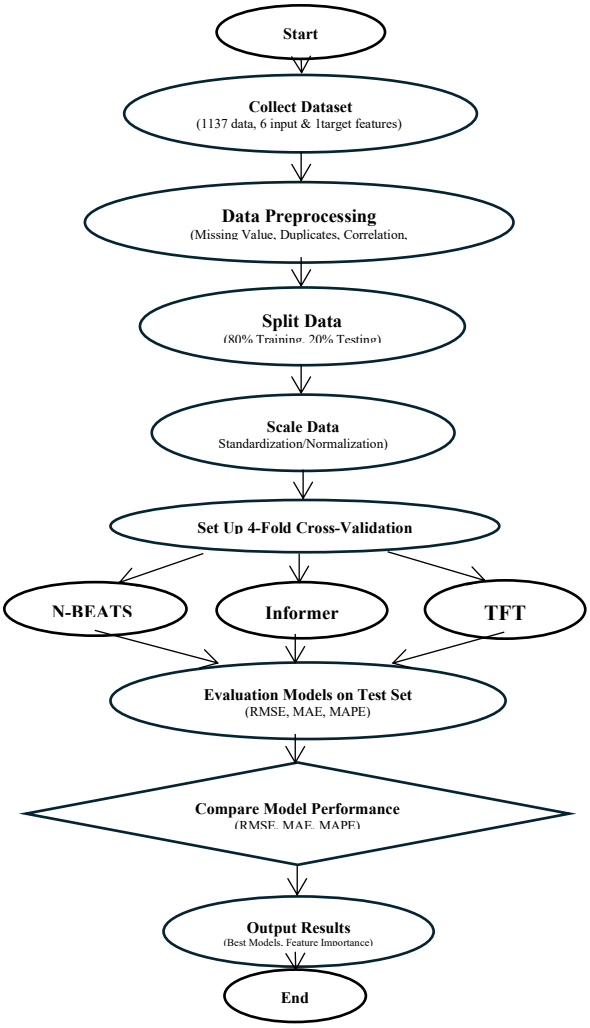


Fig. 2. Flowchart of the Machine Learning Workflow

This flowchart shows the workflow in machine learning to predict Turkish Airlines stock prices. The process begins with dataset collection (1137 data points, 6 input features, 1 target), followed by a pre-processing stage that includes handling missing values, duplicate correlation analysis, and outlier detection. The data is then divided into two parts (80% training, 20% testing), and scaling (standardization or normalization) is applied. A 4-fold cross-validation process is performed, followed by training three models, namely N-BEATS, Informer, and TFT, with hyperparameter tuning. The models were evaluated using the test dataset with the RMSE, MAE, and MAPE metrics. The performance of the models was compared based on these metrics, and the process was concluded with the selection of the best model and the identification of important features.

Exploratory Data Analysis (EDA). The Exploratory Data Analysis (EDA) stage is a very important first step in the data analysis process. The purpose of EDA is to understand the general characteristics of the dataset, discover patterns, and identify anomalies or irregularities in the data before more in-depth modeling is performed [20]. This step helps

to ensure that all elements of the data have been properly reviewed, accurately measured, and well organized. Data structure and quality checks are performed by examining missing values, duplicate data, and potential outliers.

Preprocessing. Preprocessing is an important step in data processing as it determines the quality of the input received by the prediction model. In time series research, especially in financial data such as stock prices, preprocessing aims to ensure that the data is clean, consistent, and conforms to the temporal structure required by the algorithm [20].

This step is important because deep learning models are very sensitive to noise, scale irregularities, and natural market anomalies which, if not handled properly, can hinder the learning process and reduce prediction accuracy [21]. Through preprocessing, the data is prepared to reflect the actual historical patterns without distortion, so that each variable can provide relevant information to the model.

Evaluation Metrics. The performance evaluation of the stock price prediction model is done with three main metrics, namely Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). These three metrics were chosen because they are able to measure prediction accuracy from various aspects, both in terms of absolute error, squared error, and error relative to the actual value [22].

Root Mean Squared Error (RMSE) measures the root of the average squared error between predicted and actual values. The RMSE formula is as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{9}$$

Mean Absolute Error (MAE) calculates the average absolute value of the difference between the actual and predicted values, and is given by the formula:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} | y_i - \hat{y}_i | \tag{10}$$

Mean Absolute Percentage Error (MAPE) measures the prediction error in percentage terms relative to the actual value, formulated as:

$$\text{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \tag{11}$$

The RMSE metric is more sensitive to outliers as it squares the error, whereas MAE provides a more balanced measure of error. MAPE is useful for error interpretation on a percentage scale, although it is not recommended if the data contains actual values close to zero [22].

Feature Importance Post-Hoc. Feature Importance Post Hoc (Ablation Analysis) is a technique within the Explainable AI (XAI) method that aims to provide an explanation of a previously trained model by removing, changing, or "turning off" one feature at a time, then seeing how the model's performance changes [23]. The ablation study in this research is focused on TFT because this model specifically includes external variables in its prediction process.

3 Analysis and Discussion

This chapter presents the research results obtained through analyzing the data attributes, revealing hidden trends through investigating the baseline data, and explaining the steps in model building. The purpose of this analysis is to provide a clear understanding of how the data works and to validate the selection of the prediction model used in this study.

3.1 Data Characteristics

The first important thing is to see how the data looks like. In this research, it must be able to understand the characteristics of the data.

Table 2. Descriptive Statistics of THYAO Stock Price Data

	ds	y	Close	High	Low	Open	Volume
Count	1167	1167	1167	1167	1167	1167	1167000e+03
Mean	2023-05-04	159.10	165.14	165.14	160.50	162.83	6254027e+07
Min	2021-01-04	10.64	11.23	11.23	10.72	10.86	3000000e+01
25%	2022-03-02	27.61	28.86	28.86	27.95	28.43	2.983977e+07
50%	2023-05-04	141.15	147.19	147.19	143.00	144.80	4.628326e+07
75%	2024-07-06	279.51	290.37	290.37	282.50	286.12	8.251149e+07
Max	2025-09-04	339.99	346.25	346.25	338.50	245.00	3659008e+08
std	NaN	116.98	121.07	121.07	118.28	119.74	4618980e+07

The descriptive statistics table reveals that the stock data set consists of 1,167 daily observations in the time span of January 4, 2021 to September 4, 2025, which is sufficient to analyze the time series in the medium to long term. The Adj Close (y) variable recorded an average of 159.10 with a very wide range, reflecting significant price increases as well as high levels of volatility. A similar pattern also emerges in the Open, High, Low and Close variables, where the median exceeding the lower quartile indicates that price increases dominated over the recent period. Meanwhile, the Volume variable shows huge variations, suggesting high trading dynamics. Based on these characteristics, the data exhibits non-stationary and volatile properties, which necessitates good preprocessing and a time series model that can capture complex temporal patterns. Therefore, to find out more about how the data looks like, we look at some data visualizations.

3.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the initial characteristics of THYAO's stock price data before further modeling was carried out. The data used is in the form of an adjusted closing price (Adjusted Close Price) with a time range of 2021 to 2025.

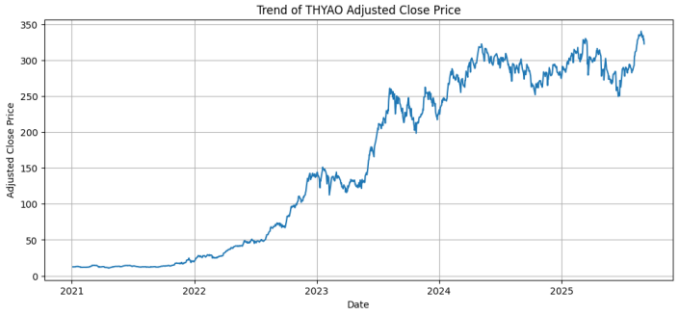


Fig. 3. Trend of THYAO Adjusted Close Price

Trend Analysis pf Adjusted Closing Price. THYAO's stock price movement shows a significant upward trend in the long run. In the early period of 2021, stock prices were relatively stable with low fluctuations. However, from mid-2022 to 2023, there has been a sharp increase in prices, which indicates a positive market sentiment towards the stock. Entering the period 2024 to 2025, although the upward trend is still continuing, price movements show higher fluctuations. This reflects increasing market volatility, which has the potential to be influenced by external factors as well as the general condition of the aviation industry.

Distribution of Adjusted Close Price. The price distribution of THYAO's stock adjusted close does not form a normal distribution pattern. The distribution of the data tends to be asymmetrical (skewed) with multiple peaks (multimodal), which indicates that stock prices are at some dominant value range in the observation period. This condition indicates a change in price phase over time, where THYAO shares were once at low, medium, to high price levels. Non-normal and heavy-tailed distributions are typical characteristics of financial asset prices and must be carefully considered when applying statistical models that assume normality [24].

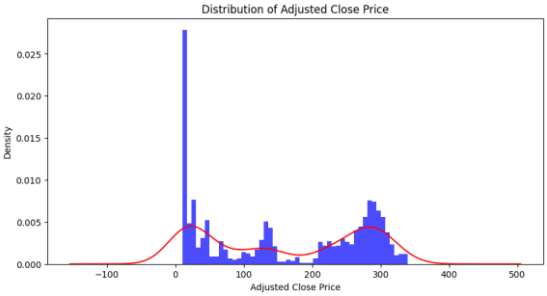


Fig. 4. Distribution of Adjusted Close Price

Trading Volume Analysis. The trading volume of THYAO shares showed considerable fluctuations in the initial observation period, particularly in 2021 to 2022. This high trading activity indicates that there is great market interest in THYAO shares in the period. Entering 2023 to 2025, trading volumes tend to be more stable even though they continue to experience spikes at certain times. This pattern suggests that price changes are not always followed by a proportionate increase in volume, which indicates that there are other factors outside of trading volume that affect stock price movements.

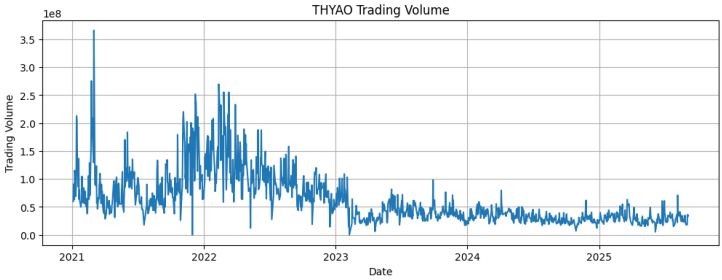


Fig. 5. THYAO Trading Volume

Based on the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots the autocorrelation values at the initial lags are statistically significant and gradually decay as the lag increases, indicating a strong temporal dependence in the data and confirming that the time series does not follow a white noise process. In contrast, the PACF plot shows a significant spike only at the first lag, while subsequent lags fall within the confidence bounds and are not statistically significant. This pattern suggests that, after controlling for the effects of previous lags, only the first lag has a direct influence on the current value. The combination of a slowly decaying ACF and a clear cut-off in the PACF at lag one indicates that the underlying data structure follows a low-order autoregressive process, supporting the use of an AR(1) component in the time series modeling framework.

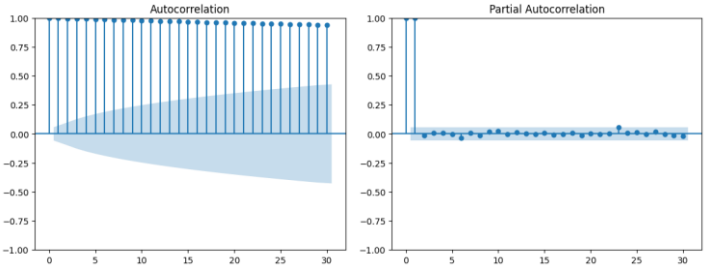


Fig. 6. Plot ACF and PACF

Average Stock Price Analysis. The Analysis of the average monthly and annual stock price, THYAO stock shows a consistent upward trend during the observation period. On a monthly level, there has been a gradual increase that has become increasingly evident since 2022, with short-term fluctuations still occurring but not changing the direction of the main trend. Meanwhile, on an annual basis, the average share price has seen a significant increase from around 13.9 in 2022 to 60.8 in 2023, then increased further to 186.5 in 2024 and reached around 297.7 in 2025. This pattern indicates a sustained growth in stock value in the long term, so that the average monthly and annual stock price can be used as an effective representation to illustrate the dynamics of THYAO's stock price trend in the context of subsequent analysis and modeling.

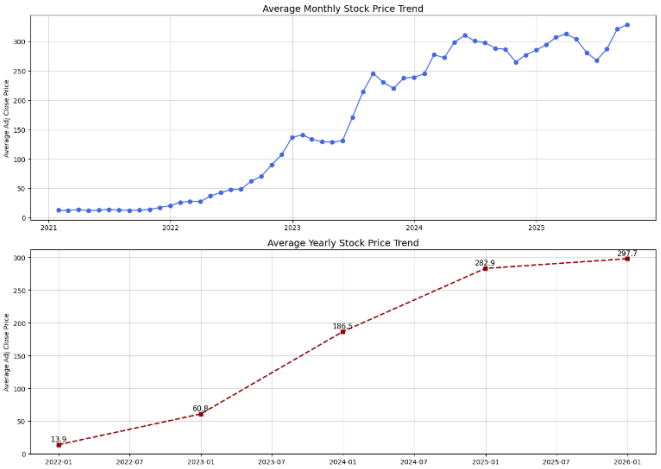


Fig. 7. Average Stock Price Monthly Trend and Yearly Trend

3.3 Hyperparameter Tuning

Table 2. Result of Hyperparameter Tuning

Input Size	Hidden Size	n_head	RMSE
20	32	2	5.925
30	32	2	6.74
20	32	4	6.993
30	64	4	7.522
30	32	4	7.756
20	64	4	7.91

Hyperparameter tuning on the TFT model was performed by varying three main hyperparameters, namely input size, hidden size, and number of attention heads (n_head). The input size parameter was tested at values of 20 and 30, which represent the length of the historical data context used as model input. The hidden size parameter was tested at values of 32 and 64, which control the dimension of the latent representation and the model's capacity to learn temporal patterns. Meanwhile, the number of attention heads parameter was tested at values of 2 and 4, which determine the number of parallel attention mechanisms in the transformer architecture. All combinations of hyperparameters were evaluated using a manual *grid-based tuning* approach, with the condition that the hidden size value must be divisible by the number of attention heads. The performance evaluation of each combination was carried out using rolling window cross-validation and the Root Mean Squared Error (RMSE) metric, then the configuration with the lowest RMSE value was selected as the optimal configuration.

3.4 Model Definition

At this stage, the architecture of the artificial neural network (ANN) designed to predict the Adjusted Close (y) stock price of Turkish Airlines (THYAO) is specified. Based on the hyperparameter tuning results in the previous section, an observation window (input size) of 20 periods was obtained. All models were trained using the Mean Absolute Error (MAE) loss function to provide robustness against extreme stock price fluctuations. A detailed explanation of the implementation of the N-BEATS, Informer, and Temporal Fusion Transformer models is as follows.

N-BEATS (Neural Basis Expansion Analysis). N-BEATS is a time series forecasting model with a pure deep learning approach that consists of stacked functional blocks in the form of fully-connected layers [18]. In this study, *stack_types=['identity']* was used to enable the model to learn non-linear mappings from the data flexibly without forcing specific trend or seasonal assumptions [18]. This approach makes N-BEATS highly adaptive to the stochastic and irregular movements of THYAO stock prices [25].

Informer. Informer is a model designed to overcome efficiency limitations in conventional Transformer architecture when handling very long sequences . In this study, Informer acts as univariate model that focuses on learning only one thing, namely the historical price of THYAO shares. The main mechanism that distinguishes Informer is the use of ProbSpars Self-attention, which is selection technique that only calculates attention scores on elements that have the highest statistical contribution to sequence representation [16]. This effectively reduces computational complexity from $O(L^2)$ to $O(L \log L)$, without sacrificing the quality of information representation [16]. Based on optimization, the model is configured with *hidden_size=32* and *n_head=2*.

Temporal Fusion Transformer (TFT). TFT is a model that uses the Transformer architecture and utilizes the self-attention mechanism to capture complex temporary changes from various time series simultaneously. This model is used for multi-horizon forecasting with transparent prediction result explanation capabilities. This model utilizes a Variable Selection Network (VSN) that functions to select the most relevant exogenous variables, which in this study include the Volume, Open, High, and Low variables [19]. Then there is the Gated Residual Network (GRN) component, which enables the model to automatically determine whether a the data requires complex non-linear processing or can be passed directly through the gating mechanism [19]. The optimal configuration in this model is *input_size=20, hidden_size=32, and n_head=2*.

3.5 Roling Cross Validation

After the Informer, Temporal Fusion Transformer (TFT), and N-BEATS models are built, the next stage is the application of the Rolling Cross Validation that applied to create a more realistic time series forecasting scenario, where each model is always built using available data and tested in future periods. In this study, the data used for evaluation is taken as 20% of the total data, which includes 228 observations, set as the number of evaluation windows (n_windows = 228), with the configuration *val_size = 10* and *step_size = 1*, so that the validation window shifts one step at each iteration. This approach allows the evaluation of model performance to be performed iteratively across successive time segments, thus providing a more robust and fair analysis than the conventional method of dividing training and testing data, especially with stock data that tends to be volatile and fluctuating.

The results of Rolling Cross Validation show that the N-BEATS, Informer, and TFT models are always retrained in each window until they reach the maximum training limit (max_steps = 500), while featuring the model structure, number of adjustable parameters, and GPU utilization, so that the performance comparison between models is based on a systematic evaluation method that reflects the real-world situation.

3.6 Model Evaluation and Comparison

Table 3. Evaluation Results

Models	MAPE	RMSE
N-BEATS	1.55%	6.0910
Informer	2.08%	7.5920
TFT	1.76%	6.6567

From table 3 (the evaluation results) show that the Informer model provides the best performance in predicting the Adjusted Close price of THYAO shares with a MAPE value of 1.55% and an RMSE of 6.0910, making it the most accurate model in the test data. The N-BEATS model ranked second with a MAPE of 1.76% and an RMSE of 6.6567, demonstrating its competitive ability to capture the nonlinear patterns of stock price time series. Meanwhile, the Temporal Fusion Transformer (TFT) produced a relatively higher error rate, with a MAPE of 2.08% and an RMSE of 7.5920, despite using exogenous variables. This indicates that adding external variables does not always improve prediction accuracy and, in the context of THYAO shares, can actually add noise, making univariate Transformer-based models such as Informer more effective in dealing with high market volatility.

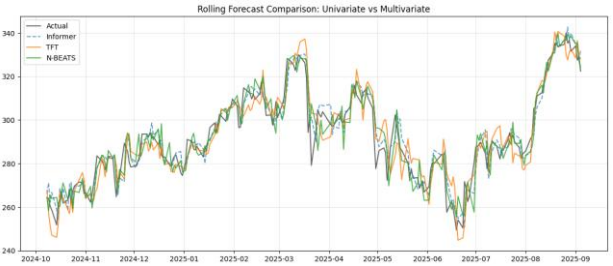


Fig. 8. Rolling Forecast Comparison: Univariate vs Multivariate

It appears that the Informer (univariate) model is the most consistent in approximating actual data in following short-term trends and fluctuations with relatively small deviations. N-BEATS is still able to follow general patterns but shows a slight lag when trends change rapidly. while TFT (multivariate) tends to over-smooth and deviate during periods of high volatility. Overall, Informer provides the best performance, followed by N-BEATS, with TFT being the least optimal.

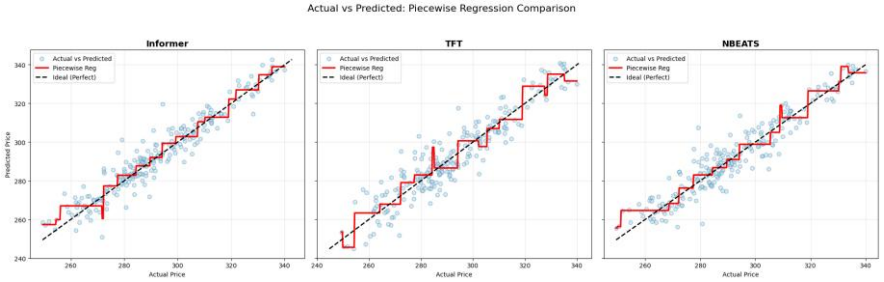


Fig. 9. Actual vs Predicted: Piecewise Regression Comparison

Visually, Informer shows the best fit because the actual and predicted points are most closely spread around the ideal line and the piecewise regression line consistently follows the diagonal across the entire price range, indicating low bias and high prediction stability. N-BEATS is still able to capture the linear relationship, but in some price segments it shows small deviations that indicate a lag, while TFT has a wider distribution of points and greater deviation from the ideal line, resulting in a relatively higher error. These visual findings are

in line with the quantitative evaluation, where based on ranking with MAPE as the main criterion and RMSE as a supporting criterion, the Informer model was selected as the best model with a MAPE of 1.55% and an RMSE of 6.0910, which shows the lowest prediction error rate compared to other models.

3.7 Post-Hoc Analysis

Post-hoc analysis is conducted with the aim of evaluating model behavior in depth to understand the quality of predictions in various market conditions. The use of Explainable AI (XAI) techniques at this stage is to understand how complex deep learning models work, so that the process of obtaining prediction figures becomes clearer, more transparent, and scientifically accountable [26].

Ablation Analysis.

This study specifically uses post-hoc ablation for TFT because its architecture is designed to explicitly adjust and process the influence of exogenous variables[19]. This method is in line with the XAI principle, which emphasizes the importance of feature evaluation to see how the model makes decisions after the training process is complete [26]. Feature contribution is measured through the masking method, where each exogenous variable is replaced with its average value in turn to calculate the difference in Mean Absolute Error (MAE) increase [25].

The formula used in this calculation is.

$$\Delta MAE = MAE_{masked} - MAE_{baseline} \tag{12}$$

The MAE metric was chosen because it measures the impact of errors directly in price units, so it can be seen how large the price error is if certain information is omitted compared to RMSE [25].

Interpretation of results.

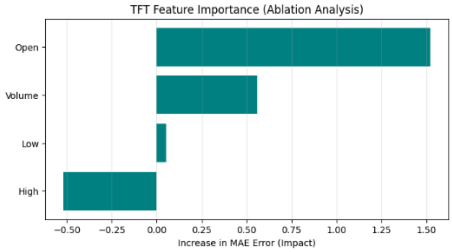


Fig. 10. TFT Feature Importance (Ablation Analysis)

Based on the ablation results, the TFT Feature Importance graph shows that the Open variable is the most influential exogenous variable on model performance, with an MAE increase of around 1.5. The second position is occupied by the Volume variable, followed by the Low variable, while the High variable becomes noise with a negative impact of around -0.5. This indicates that variable High had worsens accuracy, so that if it is removed, it has the potential to reduce the overall prediction error value. Therefore, this proves why the univariate Informer model provides better results than the multivariate TFT model, because internal price history provides a more transparent explanation of prediction results without errors from irrelevant external variables.

4 Conclusion

This study investigates how well sophisticated deep learning models, namely Informer, N-BEATS, and Temporal Fusion Transformer (TFT), can predict the Adjusted Close price for Turkish Airlines (THYAO) shares from 2021 to 2025, a period known for considerable economic instability in Turkey. The results suggest that previous price data, shown through lag variables, is very important for understanding how THYAO stock prices change over time. Of the models tested, Informer provides the most accurate predictions, scoring the lowest on MAPE and RMSE, showing its ability to handle long-term dependencies in very unstable market situations. N-BEATS performs similarly in identifying complex, nonlinear price trends, although it shows slight lags when sudden changes in trend occur. In contrast, TFT shows relatively higher error rates, implying that including outside factors like Open, High, Low, and Volume doesn't always improve prediction accuracy and might instead add unnecessary complexity. Additional ablation tests show that several outside variables offer little or even negative impact on the model, which explains why the simpler, univariate Informer outperforms the more complex, multivariate TFT. To sum up, these findings highlight how effective transformer-based univariate models are for predicting stock prices in the airline industry, which is often marked by high volatility and unpredictable behavior.

Beyond the effectiveness of the model, these results highlight the significant effect of Turkey's overall economic situation, particularly rising inflation, exchange rate fluctuations, and energy cost variations on THYAO stock behavior. These elements add unpredictable and highly unstable price variations, emphasizing the need for prediction methods that rely on previous price trends rather than unreliable external data. In this type of setting, models designed to identify patterns that are inherent over time provide more reliable and solid estimates.

5 Recommendation

Given how significantly airline stocks, such as THYAO, can be affected by major economic shifts, currency fluctuations, and energy cost volatility, Turkish policymakers and financial market participants must focus on controlling inflation and maintaining currency stability to reduce uncertainty in financial markets. From an investment perspective, evidence shows that historical price data remains a solid and informative way to predict THYAO's stock movements, even when the economy is shaky. Therefore, investment strategies should emphasize the use of forecasting models based on solid data and reliable time series methods, rather than relying too much on external factors that may only add to the confusion. Additionally, airlines are encouraged to make their financial reports and performance metrics more transparent, so that investors have greater confidence and the risk of drastic price fluctuations due to speculation is reduced.

For future investigations could expand this research by adding broader economic factors, such as price inflation rates, currency values, electricity costs, and borrowing costs using organized and focused methods, such as selecting key characteristics or methods to determine cause and effect, to avoid repetition and overly complex models. Additionally, looking at combined modeling strategies that blend transformer-based designs with models using probabilities or change patterns could improve the ability to understand shifts in monetary data structures over time. Furthermore, using AI methods that are more understandable than simply removing parts for analysis, such as looking at where the model focuses or using SHAP-based understanding tools, could provide deeper insight into how the model acts. Finally, expanding the study to include other airline stocks or more developed economic areas will help to verify whether the results are valid in many cases and reinforce the practical use of transformer-based prediction models in monetary studies.

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